SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

STUDENT TASK MANAGEMENT SYSTEM

Internship Organization: CodeOrbit Tech

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1. INTRODUCTION

1.1 Purpose

The Student Task Management System is designed to help students organize and manage their academic activities. It allows students to add, view, update and delete tasks easily.

Students regularly have multiple assignments, homeworks, projects and examination preparations. Managing all these activities manually can sometimes become difficult.

This system provides a centralized platform where students can add new tasks, view existing tasks, update task details and delete completed or unnecessary tasks.

The main purpose of this project is to improve student organization skills, save time and help them to complete their academic activities within the given deadlines.

1.2 Project Scope

This application provides a simple platform for managing academic tasks. Its main features include:

1. Adding new tasks.
2. Viewing existing tasks.
3. Updating task details.
4. Deleting unwanted tasks.
5. Setting task deadlines.
6. Tracking pending and completed tasks.

1.3 Intended Users

The system is mainly designed for:

1. College Students: To manage assignments and projects.
2. School Students: To organize homework.
3. Project Students: To track project activities and deadlines.

1.4 Definitions and Abbreviations

- SRS: Software Requirements Specification.
- UI: User Interface.
- CRUD: Create, Read, Update and Delete.
- Task: An activity that needs to be completed.
- Deadline: The date by which a task must be completed.

2. PROJECT OBJECTIVES

The main objective of the Student Task Management System is to develop a simple and efficient application that helps students organize, manage and track their academic activities. The system aims to reduce manual work, improve time management and increase student productivity.

1. Task Organization

To provide a centralized platform where students can maintain all their academic activities, including assignments, homework, practical work and projects.

2. Efficient Time Management

To help students plan their daily activities and complete their tasks within the assigned deadlines.

3. Easy Task Management

To provide simple options for adding, viewing, updating and deleting tasks without requiring advanced technical knowledge.

4. Progress Tracking

To help students identify pending and completed tasks so they can monitor their academic progress.

5. Improved Productivity

To encourage students to organize their work systematically, avoid unnecessary delays and complete their academic responsibilities on time.

6. Reduced Manual Effort

To minimize the need for maintaining task details manually and make task management easier and faster.

7. Better Planning and Prioritization

To help students organize important activities and focus on tasks that require immediate attention.

8. User-Friendly Experience

To develop an application with a simple and understandable interface that can be easily operated by students.

3. FUNCTIONAL REQUIREMENTS

Functional requirements describe the main operations that the Student Task Management System must perform.

1. Add Task: The system shall allow users to add new tasks by entering a title, description and deadline.

2. View Task: The system shall display all available tasks with their details and current status.

3. Update Task: The system shall allow users to modify existing task information whenever required.

4. Delete Task: The system shall provide an option to remove unwanted tasks after user confirmation.

5. Task Status: The system shall allow users to mark tasks as Pending or Completed.

6. Deadline Management: The system shall allow users to assign due dates to their academic tasks.

7. Input Validation: The system shall check user inputs and prevent empty task titles or invalid information.

4. NON-FUNCTIONAL REQUIREMENTS

Non-functional requirements describe the quality, performance and usability of the application.

1. Usability: The application should have a simple and user-friendly interface that students can easily understand.

2. Performance: The system should respond quickly when users perform different operations.

3. Reliability: The application should perform its functions correctly and maintain accurate task information.

4. Security: The system should protect user information and prevent unauthorized access wherever security features are implemented.

5. Maintainability: The source code should be properly organized so developers can easily update and improve the application.

6. Portability: The application should be designed to run on supported operating systems with the required software environment.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware and software requirements describe the basic resources needed to develop and run the Student Task Management System.

5.1 Hardware Requirements

The following hardware components are required:

1. Processor: Intel or equivalent processor for running the application smoothly.

2. RAM: Minimum 4 GB RAM for basic development and execution.

3. Hard Disk: At least 500 MB of available storage space for project files and software.

4. Keyboard: Required for entering task details and providing user input.

5. Mouse: Used for navigating the development environment and application.

6. Monitor: A standard display for viewing the application and its output.

5.2 Software Requirements

The following software tools are required for project development:

1. Operating System: Windows, Linux or macOS.

2. Programming Language: Java, used to develop the application.

3. Development Environment: Visual Studio Code (VS Code), used for writing and editing Java programs.

4. Java Development Kit (JDK): Required to compile and execute Java programs.

5. Documentation Tool: Google Docs, used to prepare the Software Requirements Specification document.

6. Version Control: Git and GitHub, used to manage and store project source code.

5.3 Development Environment

The application will be developed using Java programming language in Visual Studio Code. The JDK will provide the necessary tools to compile and execute the program. GitHub will be used to maintain the source code and track project changes.

6. USE CASES

Use cases describe how students interact with the Student Task Management System to perform different operations. They explain the steps followed by users to complete specific tasks.

6.1 Add Task

Actor: Student

Description: Allows students to add new academic tasks.

Steps:

1. The student opens the application.
2. Select the Add Task option.
3. Enter the task title, description and deadline.
4. The system validates the entered information.
5. The task is added to the task list.

Expected Result: The new task is successfully saved.

6.2 View Tasks

Actor: Student

Description: Allows students to view their existing tasks.

Steps:

1. The student opens the application.
2. Select the View Tasks option.
3. The system retrieves the available tasks.
4. All task details are displayed.

Expected Result: The student can view all available tasks.

6.3 Update Task

Actor: Student

Description: Allows students to modify existing task details.

Steps:

1. Select the Update Task option.
2. Enter the task identification number.
3. The system finds the selected task.
4. The student modifies the required details.
5. The system saves the changes.

Expected Result: The selected task is updated successfully.

6.4 Delete Task

Actor: Student

Description: Allows students to remove unwanted tasks.

Steps:

1. Select the Delete Task option.
2. Enter the task identification number.
3. The system displays the selected task.
4. The student confirms the deletion.
5. The system removes the task.

Expected Result: The selected task is deleted successfully.

6.5 Complete Task

Actor: Student

Description: Allows students to mark tasks as completed.

Steps:

1. Open the task list.
2. Selects the required task.
3. Changes its status to Completed.
4. The system updates the task status.

Expected Result: The task is marked as completed.

7. SYSTEM OVERVIEW AND WORKFLOW

7.1 System Overview

The Student Task Management System is designed to help students organize and manage their academic activities efficiently. It provides a simple platform for creating, viewing, updating and deleting tasks.

The application accepts information from users, processes their requests and displays the required results. Each task contains important details such as task title, description, deadline and completion status.

The main purpose of the system is to make academic task management easier and help students keep track of their daily activities.

7.2 System Workflow

The system follows a simple sequence of operations:

1. **Start Application:** The user opens the Student Task Management System.
2. **Display Menu:** The application displays available options such as Add, View, Update and Delete Task.
3. **User Selection:** The user selects the required operation.
4. **Input Details:** The system accepts the necessary information from the user.
5. **Input Validation:** The application checks whether the entered information is valid.
6. **Process Request:** The system performs the selected operation.
7. **Display Result:** The application displays a success message or an appropriate error message.
8. **Continue or Exit:** The user can perform another operation or exit the application.

7.3 Main System Modules

1. Task Creation Module: Allows users to add new academic tasks with the required details.

2. Task Display Module: Displays all available tasks and their information.

3. Task Update Module: Allows users to modify existing task details.

4. Task Deletion Module: Removes unwanted tasks from the system.

5. Task Status Module: Helps users identify pending and completed tasks.

8. ASSUMPTIONS AND LIMITATIONS

8.1 Assumptions

1. Users have basic knowledge of operating a computer.
2. Users provide correct task details and deadlines.
3. Each task is assigned a unique identification number.
4. The application is mainly intended for individual student use.
5. Users have the required software environment to run the application.

8.2 Limitations

1. The basic application focuses only on academic task management.
2. Automatic reminders and notifications are not included.
3. Online synchronization is not available in the basic version.
4. Advanced reports and analytics are not provided.
5. The available features depend on the implemented application.

9. CONCLUSION

The Student Task Management System is designed to help students organize and manage their academic activities efficiently. It provides essential features such as adding, viewing, updating and deleting tasks.

The application helps students track their deadlines, identify pending activities and monitor completed tasks. It also encourages better time management and reduces the effort required to maintain academic records manually.

Through this project, important software development concepts such as requirement analysis, functional requirements, non-functional requirements and system workflow are considered.

In conclusion, the Student Task Management System provides a simple and organized approach to academic task management. It can be further improved by introducing advanced features such as reminders, notifications and progress tracking.